

A PROJECT REPORT ON

**CIPHERDOCS: A SECURE AI-POWERED SOLUTION
TO PREVENT FORGERY OF CRITICAL AND
OFFICIAL DOCUMENTS USING POLYGON
BLOCKCHAIN-BASED ENCRYPTION
VERIFICATION**

**IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE
OF**

**BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY**

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[An Autonomous Institute Affiliated to Savitribai Phule Pune University]

2025-26

FRONT PAGE

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2025-26**

CERTIFICATE

This is to certify that the project report entitled "CipherDocs: A Secure AI-Powered Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption Verification" is a bona fide work carried out by Harshvardhan Nerpagar (122B1F088), Ojas Patil (122B1F093), Tejas Pokale (122B1F102), and Nisha Chavan (123B2F141), students of Bachelor of Technology in Information Technology, in partial fulfillment for the award of degree under Savitribai Phule Pune University during the academic year 2025-26.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
DBMS	Database Management System
GUI	Graphical User Interface
HTTP	HyperText Transfer Protocol
ML	Machine Learning
SDG	Sustainable Development Goals
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator

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ABSTRACT

Maintaining the integrity and authenticity of digital documents is critical as forgery and unauthorized modifications pose significant risks across education, governance, and legal domains. Existing verification systems are often centralized, inefficient, and vulnerable to tampering. This project presents CipherDocs, a decentralized document verification platform leveraging blockchain technology to ensure secure and transparent validation.

The system stores encrypted documents in IPFS while recording their SHA-256 hashes on the Polygon blockchain, ensuring integrity and immutability. AES-256 encryption provides confidentiality, and dynamic QR codes enable real-time verification. Smart contracts automate lifecycle operations such as issuance, verification, revocation, and expiry tracking.

Additionally, AI-assisted validation detects structural anomalies, and Zero-Knowledge Proof mechanisms support privacy-preserving verification. The system achieves improved efficiency, scalability, and reduced transaction costs, offering a reliable and cost-effective solution for secure digital document management.

CHAPTER 1

INTRODUCTION

1.1 Background

The rapid growth of digital documents in sectors such as education, governance, and legal systems has significantly increased the need for secure and reliable document management systems. However, this growth has also led to a rise in document forgery, unauthorized alterations, and lack of trust in verification processes. Traditional document verification systems are primarily centralized and rely on manual validation or database-based checks, making them inefficient, time-consuming, and vulnerable to single points of failure and security breaches.

Although blockchain-based solutions have been introduced to address these challenges, existing systems still suffer from several limitations such as high transaction costs, lack of dynamic verification mechanisms, limited lifecycle management, and inadequate support for advanced forgery detection. These limitations restrict their scalability and real-world applicability.

To overcome these challenges, the proposed system **CipherDocs** integrates blockchain technology, cryptographic security, decentralized storage, and intelligent validation into a unified framework. The system utilizes the **Polygon blockchain** to store cryptographic hashes of documents in a cost-efficient and scalable manner, while actual documents are stored securely in the **InterPlanetary File System (IPFS)**. Cryptographic techniques such as **SHA-256** hashing ensure document integrity, while **AES-256** encryption guarantees confidentiality.

Additionally, the system incorporates **dynamic QR code-based verification** for real-time validation of documents. Smart contracts are used to automate lifecycle operations such as issuance, verification, revocation, and expiry tracking. Furthermore, **AI-assisted validation** is integrated to detect structural and textual anomalies, enhancing the system's ability to identify forged or tampered documents beyond traditional hash comparison.

The proposed system provides a secure, decentralized, and scalable solution for document verification and is suitable for applications in academic certification, government records, and legal documentation, thereby improving transparency, trust, and operational efficiency.

1.2 Motivation

The key motivation is to design a system that ensures **document security, authenticity, and transparency** without relying on centralized authorities. The project is driven by the following practical concerns:

- Increasing cases of document forgery and unauthorized modification in digital systems.

- Lack of trust in traditional verification methods due to centralized control.
- Inefficiency and delays in manual or semi-automated verification processes.
- Risk of data tampering, single point failures, and exposure of sensitive information.
- Need for a decentralized, scalable, and tamper-proof verification mechanism using blockchain.

1.3 Aims and Objectives

1. Build a decentralized document verification system using blockchain technology.
2. Ensure document integrity using cryptographic hashing (SHA-256).
3. Provide confidentiality through encryption techniques such as AES-256.
4. Implement decentralized storage using IPFS for secure document handling.
5. Enable real time verification using dynamic QR codes.
6. Develop smart contract based lifecycle management including issuance, verification, revocation, and expiry.
7. Integrate AI assisted validation to detect structural and textual anomalies in documents.
8. Design a user friendly interface for document upload and verification.

1.4 Sustainable Development Goals (SDG) Addressed

The project aligns with the following SDGs:

- **SDG 9 (Industry, Innovation and Infrastructure):** by developing secure and innovative digital verification infrastructure.
- **SDG 10 (Peace, Justice and Strong Institutions):** by improving transparency, trust, and authenticity in document verification systems.
- **SDG 16 (Reduced Inequalities):** by providing accessible and reliable verification tools for all users.
- **SDG 17 (Partnerships for the Goals):** by enabling collaboration between institutions, governments, and technology platforms.

1.5 Organization of Project Report

Chapter 1 introduces the problem context, motivation, objectives, and SDG mapping. Chapter 2 reviews literature on blockchain, cryptographic security, and document verification systems. Chapter 3 defines functional and non functional requirements along with tools and technologies used. Chapter 4 describes project planning, timeline, and cost estimation. Chapter 5 explains the methodology and system architecture.

Chapter 6 presents system design including use case diagrams, activity diagrams, sequence diagrams, ER diagrams, and deployment architecture. Chapter 7 details the implementation of modules such as blockchain integration, encryption, IPFS storage, and QR based verification. Chapter 8 presents testing strategies and results. Chapter 9 discusses outcomes and comparative analysis. Chapter 10 concludes the work and suggests future enhancements.

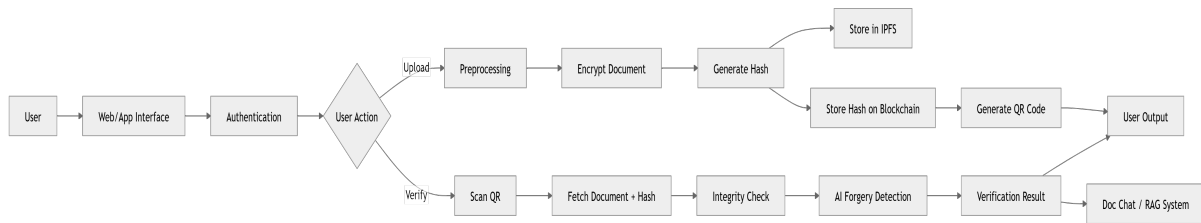


Figure 1.1. CipherDocs Conceptual Workflow

CHAPTER 2

LITERATURE REVIEW

This chapter presents a review of existing research related to blockchain-based document verification, forgery detection, and decentralized storage systems. The analysis is based on selected papers focusing on security, immutability, cost efficiency, and practical deployment of document verification systems.

2.1 Review Method

The literature survey was conducted using research papers from the project reference set and recent publications in blockchain-based certification systems.

The selection criteria included:

- Relevance to document verification and forgery prevention.
- Use of blockchain technology for immutability and transparency.
- Integration of cryptographic techniques such as hashing and encryption.
- Use of decentralized storage systems like IPFS.
- Practical feasibility and scalability for real-world deployment.

The comparison parameters include security, transaction cost, scalability, privacy preservation, and system efficiency.

2.2 Paper-Wise Analysis

No.	Title/Author	Key Contribution	Limitation
1	Certifier DApp: Decentralized Certification System (Jha et al., 2023)	Ethereum-based certification with immutable records	High gas fees and manual intervention
2	Comparative Study of Ethereum and Polygon (Aung et al., 2024)	Demonstrates Polygon for faster and cheaper transactions	Limited to NFT-based certification systems

3	Systematic Review on Blockchain Certificate Verification (Rustemi et al., 2023)	Analysis of 34 systems focusing on security	Poor usability and high implementation cost
4	Decentralized Certificate Issuance System (Sree et al., 2025)	Smart contract-based automation	Performance degradation under load
5	ShikkhaChain Credential System (Farabi et al., 2025)	Region-specific blockchain credential system	Limited scalability and no forgery detection
6	DocCert Blockchain Solution (Aldwairi et al., 2023)	Document verification and authenticity using blockchain	High cost and limited scope
7	Paperless Document Systems Review (Precht et al., 2026)	Uses IPFS, hashing, and smart contracts	Limited lifecycle automation
8	Blockchain + IPFS + QR Code System (Tengse et al., 2025)	Secure storage with QR-based verification	Static QR codes and basic validation
9	Document Forgery Detection Review (Sukhija et al., 2025)	ML-based forgery detection techniques	Traditional methods fail after compression, no real-time validation
10	Blockchain Framework for Document Verification (Shettar et al., 2022)	Smart contracts with immutable records	High cost and limited privacy

2.3 Research Gap

From the literature, it is observed that existing systems provide strong individual solutions such as blockchain-based immutability, decentralized storage, or machine learning-based forgery detection. However, they lack an integrated approach that combines all these features into a single efficient system.

The major gaps identified are:

- High transaction costs in Ethereum-based systems.
- Lack of dynamic and real-time verification mechanisms.
- Absence of automated document lifecycle management.
- Limited privacy-preserving verification techniques.
- Weak integration of AI-based forgery detection with blockchain.

2.4 Proposed Solution Perspective

To address these limitations, the proposed system **CipherDocs** introduces:

- Cost-efficient blockchain implementation using Polygon.
- Secure document storage using IPFS with AES-256 encryption.
- Integrity verification using SHA-256 cryptographic hashing.
- Dynamic QR-based real-time verification.
- Smart contract-based lifecycle management (issue, revoke, expiry).
- Optional AI-assisted document validation.

This approach ensures improved security, scalability, cost efficiency, and usability compared to existing systems.

CHAPTER 3

REQUIREMENTS

3.1 Functional Requirements

The proposed system **CipherDocs** shall provide the following functionalities:

- Allow authorized issuers to upload digital documents through a web-based interface.
- Validate uploaded documents based on format and size constraints.
- Generate a cryptographic hash (SHA-256) for each document to ensure integrity.
- Encrypt documents using AES-256 before storing them in decentralized storage (IPFS).
- Store document hash and metadata on the Polygon blockchain using smart contracts.
- Generate dynamic QR codes for each document to enable real-time verification.
- Provide document verification by comparing recomputed hash with blockchain-stored hash.
- Support certificate lifecycle management including issuance, revocation, and expiry tracking.
- Enable role-based access control for issuers, owners, and verifiers.
- Provide AI-assisted validation to detect structural and textual anomalies in documents.
- Display verification results indicating document status (valid, tampered, revoked, expired).

3.2 Non-Functional Requirements

- **Security:** Ensure strong encryption (AES-256) and hashing (SHA-256) to protect confidentiality and integrity of documents.
- **Scalability:** System should handle large volumes of document transactions efficiently using Polygon blockchain.
- **Performance:** Provide fast verification with low latency and minimal transaction cost.
- **Reliability:** Ensure consistent system operation with fault tolerance in decentralized storage and blockchain interaction.

- **Usability:** Provide an intuitive and user-friendly interface for issuers and verifiers.
- **Privacy:** Ensure that sensitive document data is not exposed during verification (ZKP-ready design).
- **Maintainability:** Modular system design allowing easy updates and integration of new features.
- **Auditability:** Maintain immutable and transparent transaction logs using blockchain.

3.3 Tools and Technologies Used

- Programming Language(s): JavaScript, Solidity, Python
- Framework(s): Web3.js / Ethers.js, React (Frontend), Node.js (Backend)
- Blockchain Platform: Polygon (Proof of Stake)
- Smart Contracts: Solidity
- Storage: IPFS (InterPlanetary File System)
- Cryptography: SHA-256 hashing, AES-256 encryption
- Wallet Integration: MetaMask (Web3 authentication)
- AI Tools (Optional): Python-based ML libraries for document validation
- IDE/Editor: Visual Studio Code
- Version Control: Git and GitHub

Table 3.1. Technology Stack Summary

Component	Selected Technology
Frontend	React-based Web Interface
Backend	Node.js with Web3 integration
Blockchain	Polygon (PoS) with Solidity smart contracts
Storage	IPFS for decentralized document storage
Security	AES-256 encryption and SHA-256 hashing
Authentication	MetaMask wallet-based authentication
AI Module	Python-based document validation (optional)

CHAPTER 4

PROJECT PLAN

4.1 Project Phases and Timeline

The development of the **CipherDocs** system follows a structured and phased approach to ensure systematic implementation and validation.

- **Phase 1 (Research and Planning):** Problem identification, requirement analysis, literature review, and feasibility study of blockchain-based document verification.
- **Phase 2 (System Design):** Design of system architecture including blockchain layer, IPFS storage, encryption model, and smart contract structure.
- **Phase 3 (Development):** Implementation of frontend (web interface), backend services, smart contracts on Polygon, and IPFS integration.
- **Phase 4 (Integration and Testing):** Integration of blockchain, storage, and AI modules followed by functional and performance testing.
- **Phase 5 (Deployment and Documentation):** Final deployment on testnet, documentation, report preparation, and validation.

4.1.1 Timeline (Academic Schedule)

- Jan – Feb 2026: Research and System Design
- Feb – Mar 2026: Development Phase
- Mar – Apr 2026: Integration and Testing
- Apr 2026: Documentation and Final Submission

4.1.2 Submission Milestones

- Soft copy submission and guide approval: **31 March 2026**
- Mandatory research publication submission as per academic requirement
- Final report submission after approval and compliance verification

4.2 Cost Estimation

The CipherDocs system is designed to be cost-efficient by utilizing open-source tools, decentralized technologies, and free-tier services.

4.2.1 Project Size Estimation

The project includes frontend, backend, blockchain smart contracts, and AI integration modules. The estimated size is approximately **10–15 KLOC**.

4.2.2 Effort Estimation Using COCOMO Model

Using the Organic COCOMO model with $KLOC = 12$:

$$Effort = 2.4 \times (12)^{1.05} \approx 32 \text{ person-months}$$

$$Schedule = 2.5 \times (32)^{0.38} \approx 9.5 \text{ months}$$

Due to parallel development and academic constraints, the actual development duration is optimized.

4.2.3 Cost Categories

Hardware Cost Estimation

The project primarily uses existing hardware such as laptops and local machines. Additional compute resources such as cloud VMs or Google Colab are used when required.

Software Cost Estimation

Most tools used are open-source including Node.js, React, Solidity, and IPFS. AI APIs and blockchain testnet services are available under free-tier usage.

Development Cost Estimation

Development effort includes frontend, backend, blockchain, AI integration, and testing. Institutional support reduces actual monetary cost.

Hosting and Deployment Costs

Deployment is performed on local systems and free-tier cloud services. Polygon testnet eliminates transaction costs.

Table 4.1. Project Cost Estimation

Category	Resource	Estimated Cost (INR)
Hardware	Laptop / VM	Existing (No cost)
Compute Resources	Google Colab / APIs	Free Tier
AI Models	Mistral, LLaMA, OCR	Free (API-based)
Storage	IPFS	Free
Blockchain	Polygon Testnet	Free
Wallet	MetaMask	Free

Table 4.2. Software Cost Summary

Component	Technology Used	Cost
Frontend	Next.js	Free
Backend	Node.js	Free
Database	MongoDB	Free / Usage-based
Blockchain	Polygon	Free (Testnet)
Smart Contracts	Solidity	Free
AI Services	TogetherAI / Cerebras APIs	Free Tier
Storage	IPFS	Free
Hosting	Local / Cloud	Free Tier

4.3 Sustainability Assessment

4.3.1 Environmental Sustainability

The system promotes a paperless approach by digitizing document verification. It reduces physical storage, printing, and transportation needs, contributing to eco-friendly operations.

4.3.2 Economic Sustainability

CipherDocs minimizes operational costs by using decentralized infrastructure and open-source tools. It reduces losses due to document forgery and supports scalable deployment across institutions.

4.3.3 Social Sustainability

The system enhances trust and transparency by providing tamper-proof verification. It is accessible globally and reduces dependency on centralized authorities.

4.4 Complexity Assessment

4.4.1 Computational Complexity

AI-based modules such as document analysis and OCR introduce moderate computational overhead, while blockchain operations remain lightweight and efficient.

4.4.2 Algorithmic Complexity

The system integrates:

- Cryptographic hashing (SHA-256)
- Encryption (AES-256)
- Smart contract logic
- AI-based document validation

4.4.3 Implementation Complexity

The project involves integration of multiple technologies:

- Frontend: Next.js
- Backend: Node.js
- Blockchain: Polygon
- Storage: IPFS
- AI Services: External APIs

This multi-layer integration increases complexity but ensures a robust and scalable system.

CHAPTER 5

METHODOLOGY

5.1 Introduction

The CipherDocs system follows a decentralized and security-driven methodology that integrates blockchain technology, cryptographic mechanisms, and artificial intelligence to ensure secure and reliable document verification.

The approach is designed to overcome limitations of traditional centralized systems by providing immutability, transparency, and automated verification. The methodology combines blockchain-based storage, decentralized file systems, and intelligent document analysis to achieve a scalable and efficient solution.

5.2 System Architecture

The CipherDocs architecture is based on a layered and modular design consisting of the following components:

- **User Interface Layer:** Web-based interface for document upload, verification, and interaction with the system.
- **Application Layer:** Handles business logic, document processing, and communication between components.
- **Blockchain Layer:** Stores document hashes and metadata using smart contracts on the Polygon network.
- **Storage Layer:** Uses IPFS to store encrypted documents in a decentralized manner.
- **AI Layer:** Performs document analysis and anomaly detection using machine learning techniques.

5.2.1 System Workflow

The overall workflow of the system is as follows:

1. User uploads a document through the web interface.
2. The system validates the document format and size.

3. The document is encrypted using AES-256 encryption.
4. A cryptographic hash (SHA-256) of the document is generated.
5. The encrypted document is stored in IPFS, generating a unique Content Identifier (CID).
6. The document hash and metadata are stored on the Polygon blockchain using smart contracts.
7. A dynamic QR code is generated for real-time verification.
8. During verification, the document hash is recomputed and compared with the blockchain record.
9. AI-based validation analyzes document structure and content for anomaly detection.
10. The system displays the verification result (Authentic, Tampered, Revoked, Expired).

5.3 Mathematical Model

Let the document be represented as:

$$D = \{d_1, d_2, d_3, \dots, d_n\}$$

The hash of the document is computed as:

$$h = H(D)$$

where H represents the SHA-256 hashing function.

The blockchain transaction is represented as:

$$T = (ID, h, t, A_o)$$

where:

- ID = Document Identifier
- h = Document Hash
- t = Timestamp
- A_o = Owner Address

Before storage, the document is encrypted:

$$C = AES256(D, K)$$

During verification:

$$h' = H(D')$$

Verification condition:

$$V(D') = \begin{cases} 1, & \text{if } h' = h \\ 0, & \text{if } h' \neq h \end{cases}$$

5.4 Algorithm

5.4.1 Document Upload and Storage Algorithm

Input: Document D
 Validate D
 Encrypt D using AES-256 → C
 Generate hash h = SHA-256(D)
 Store C in IPFS → CID
 Store (h, CID, metadata) on blockchain
 Generate QR code
 Output: Success

5.4.2 Document Verification Algorithm

Input: Document D'
 Compute hash h' = SHA-256(D')
 Fetch stored hash h from blockchain
 If h' == h:
 status = "Authentic"
 Else:
 status = "Tampered"
 Return status

5.4.3 AI-Based Validation Algorithm

Input: Document D

Extract text using OCR
 Perform semantic analysis
 Detect anomalies or inconsistencies
 Generate trust score
 Output: AI validation result

5.5 System Features Integration

The methodology integrates multiple advanced features:

- **Blockchain Integration:** Ensures immutability and transparency.
- **Cryptographic Security:** SHA-256 for integrity and AES-256 for confidentiality.
- **Decentralized Storage:** IPFS for secure and distributed document storage.
- **Smart Contracts:** Automate document lifecycle operations.
- **QR-Based Verification:** Enables real-time validation.
- **AI-Based Analysis:** Enhances detection of document forgery beyond hash comparison.
- **ZKP Support:** Enables privacy-preserving verification.

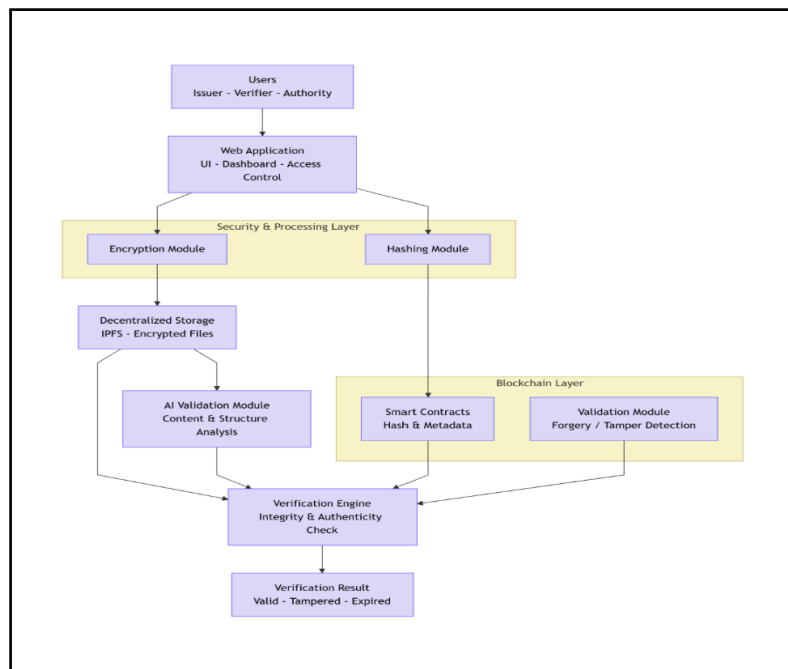


Figure 5.1. CipherDocs System Architecture

CHAPTER 6

DESIGN

6.1 Use Case Diagram

The primary actors in the CipherDocs system are:

- **Issuer:** Uploads and manages documents.
- **Holder:** Owns and accesses issued documents.
- **Verifier:** Verifies document authenticity.
- **System:** Handles blockchain, storage, and AI processing.

Key use cases include document upload, encryption, blockchain registration, verification, QR scanning, and lifecycle management.

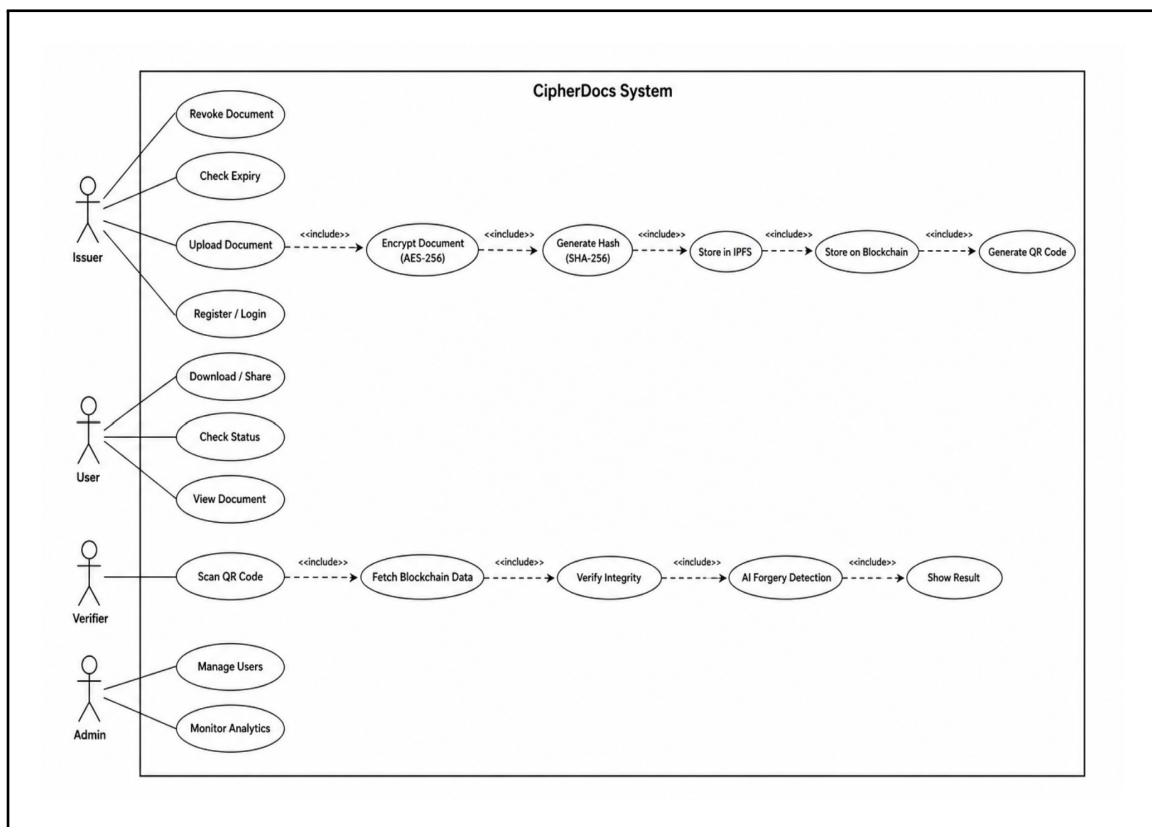


Figure 6.1. Use Case Diagram

Use Case Description:

- **Preconditions:** User authentication completed, system connected to blockchain and IPFS.
- **Main Flow:** document upload → encryption → hash generation → blockchain storage → verification.
- **Alternative Flow:** upload failure, blockchain failure, or invalid document format.
- **Postconditions:** document securely stored, hash recorded, verification enabled.

6.2 Class Diagram

The class structure of CipherDocs consists of the following major components:

- **User:** Attributes include userID, role, wallet address.
- **Document:** Contains documentID, hash, CID, status, timestamp.
- **SmartContract:** Handles issue, verify, revoke, and expiry functions.
- **IPFSStorage:** Manages encrypted document storage and retrieval.
- **BlockchainService:** Interacts with Polygon network.
- **AIValidator:** Performs document analysis and anomaly detection.

Relationships:

- User uploads Document
- Document is processed by SmartContract
- Document stored via IPFSStorage
- BlockchainService records hash
- AIValidator validates document content

6.3 Sequence Diagram

The sequence flow for document verification is as follows:

1. User uploads document via UI.

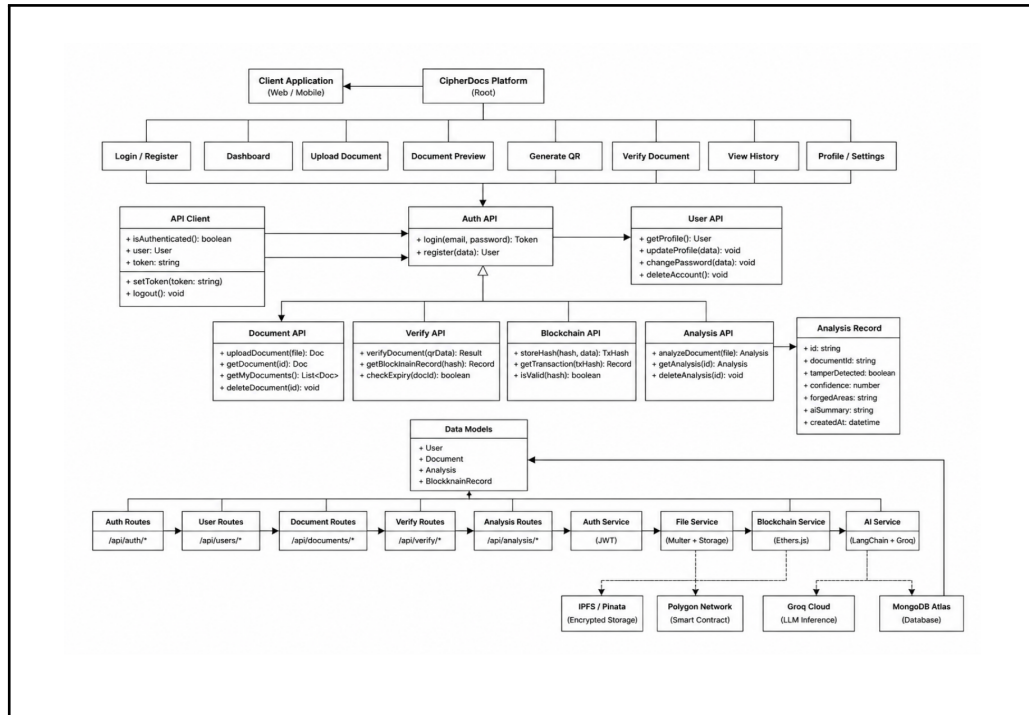


Figure 6.2. Class Diagram of CipherDocs

2. System encrypts and generates hash.
3. Document stored in IPFS.
4. Hash stored on blockchain via smart contract.
5. Verifier requests validation.
6. System recomputes hash and compares with blockchain.
7. Result is displayed to the user.

6.4 Activity Diagram

The activity flow of the system includes:

- Document upload and validation
- Encryption and hashing
- IPFS storage
- Blockchain registration
- QR code generation

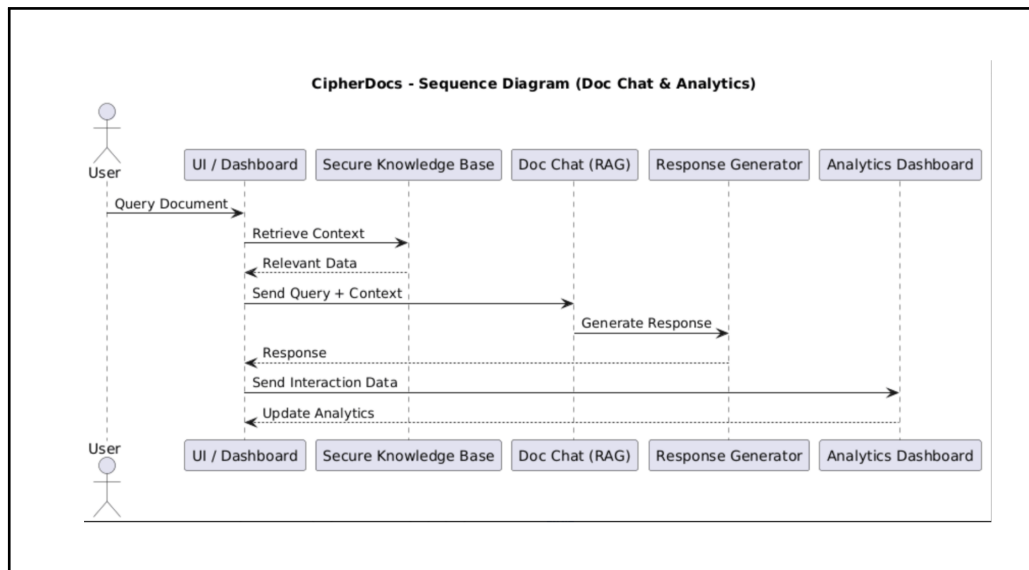


Figure 6.3. Sequence Diagram

- Verification request
- Result display (Valid / Tampered / Revoked / Expired)

Decision points:

- Valid document format?
- Hash match?
- Certificate status?

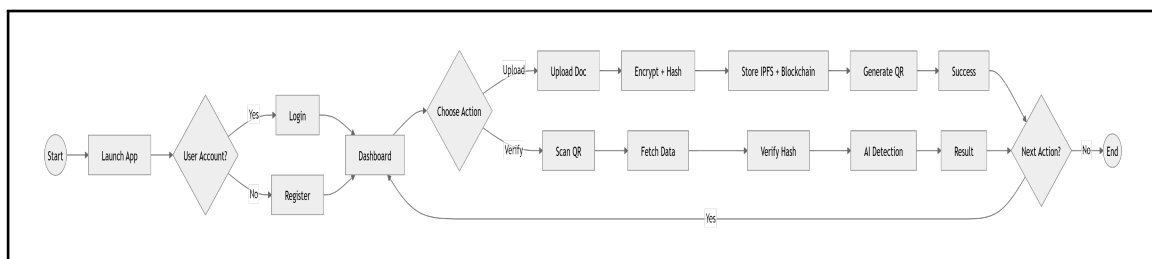


Figure 6.4. Activity Diagram

6.5 Entity-Relationship (ER) Diagram

The system uses a document-oriented structure instead of a traditional relational database.

Entities include:

- **User:** userID, role, authentication details

- **Document:** documentID, hash, CID, metadata
- **Transaction:** blockchain record containing hash, timestamp
- **Verification:** result, status, timestamp

Relationships:

- User uploads Document
- Document linked to Blockchain Transaction
- Verification references Document and Transaction

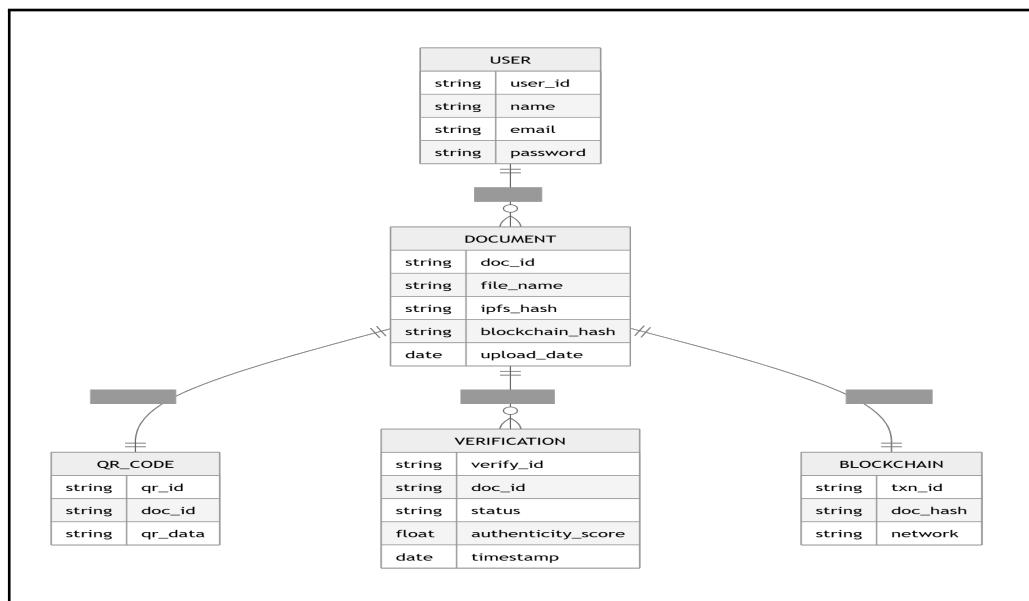


Figure 6.5. Entity-Relationship (ER) Diagram

6.6 Component Diagram

The system is divided into modular components:

- Frontend (React/Next.js UI)
- Backend (Node.js APIs)
- Blockchain Layer (Polygon + Smart Contracts)
- Storage Layer (IPFS)
- AI Module (OCR + NLP + Validation)

All components interact through APIs and Web3 interfaces.

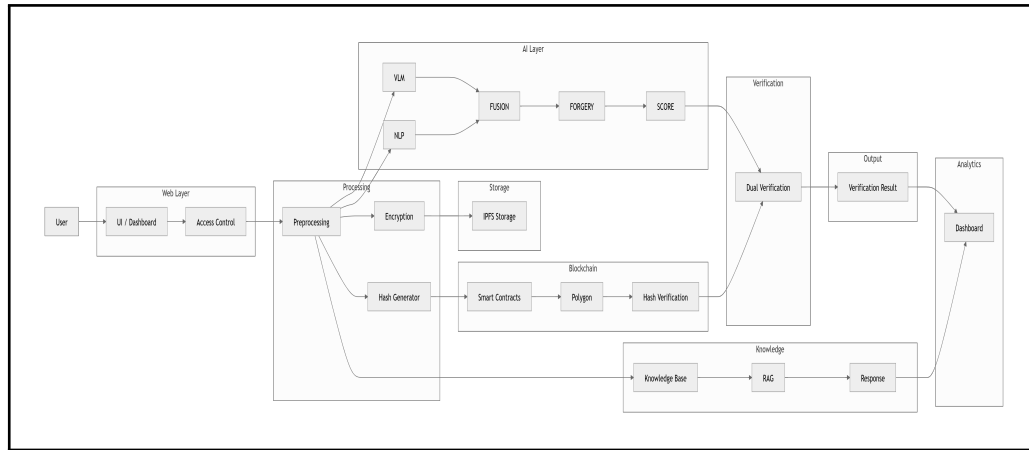


Figure 6.6. Component Diagram of CipherDocs

6.7 Deployment Diagram

The deployment architecture consists of:

- Client Device (Browser-based interface)
- Web Server (Frontend + Backend)
- Blockchain Network (Polygon)
- IPFS Network (Decentralized Storage)
- AI Services (External APIs)

The system operates in a distributed environment where blockchain ensures immutability and IPFS ensures storage resilience.

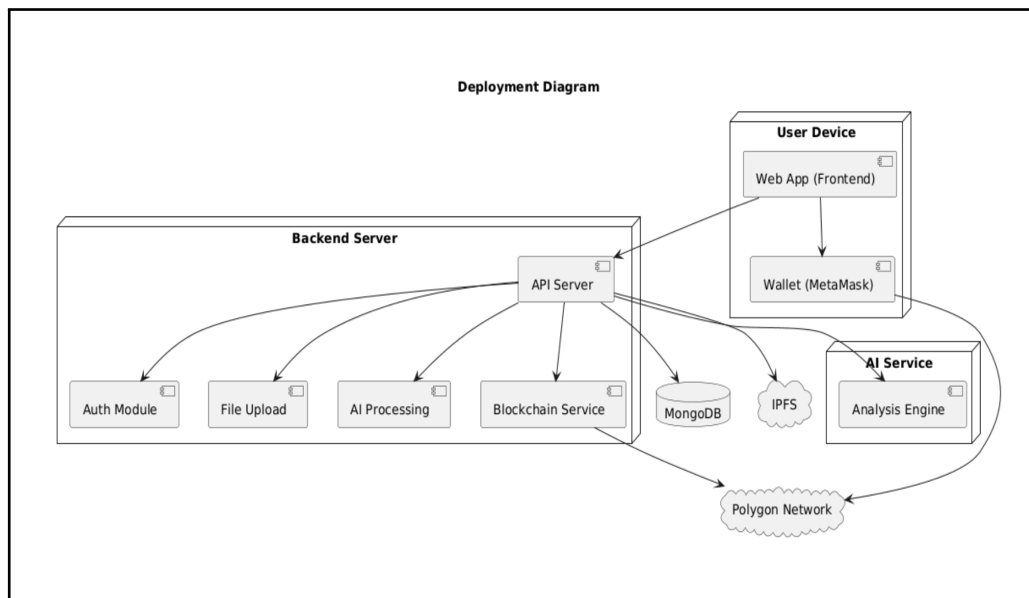


Figure 6.7. Deployment Diagram of CIPHERDOCS

CHAPTER 7

IMPLEMENTATION

7.1 Module-wise Implementation

The CipherDocs system is implemented using a modular architecture to ensure scalability, maintainability, and efficient integration of multiple technologies.

- **Frontend Module:** Developed using React/Next.js, it provides user interfaces for document upload, verification, QR scanning, and dashboard interaction.
- **Backend Module:** Implemented using Node.js and Express, it handles API requests, authentication, document processing, and communication between system components.
- **Blockchain Module:** Smart contracts written in Solidity are deployed on the Polygon network to store document hashes, metadata, and manage lifecycle operations (issue, verify, revoke, expiry).
- **Storage Module:** IPFS is used to store encrypted documents. Each uploaded document generates a unique Content Identifier (CID) for retrieval.
- **Cryptography Module:** Implements SHA-256 hashing for integrity and AES-256 encryption for secure document storage.
- **AI Module:** Uses OCR and NLP techniques for document analysis, anomaly detection, and trust score generation.
- **Authentication Module:** Uses Web3 wallet integration (MetaMask) for secure user authentication and role-based access control.

7.2 Code Snippets and Explanation

7.2.1 Document Upload and Blockchain Storage

```
function uploadDocument(file) {  
    validate(file);  
  
    const hash = SHA256(file);  
    const encryptedFile = AES256.encrypt(file);
```

```
const cid = IPFS.upload(encryptedFile);

smartContract.issueCertificate({
  documentHash: hash,
  cid: cid,
  timestamp: currentTime(),
  owner: userWalletAddress
});

return "Upload Successful";
}
```

Explanation:

- The document is first validated for format and size.
- A SHA-256 hash is generated to ensure integrity.
- The document is encrypted using AES-256.
- The encrypted document is stored in IPFS, generating a CID.
- The hash and metadata are stored on the blockchain using smart contracts.

7.2.2 Document Verification

```
function verifyDocument(file) {
  const hashComputed = SHA256(file);
  const storedHash = smartContract.getHash(fileID);

  if (hashComputed === storedHash) {
    return "Authentic";
  } else {
    return "Tampered";
  }
}
```

Explanation:

- The uploaded document is hashed again.
- The stored hash is fetched from the blockchain.
- If both hashes match, the document is authentic; otherwise, it is tampered.

7.2.3 QR Code-Based Verification

```
function generateQRCode(documentID) {  
    const verificationURL = baseUrl + "/verify/" + documentID;  
    return QRCode.generate(verificationURL);  
}
```

Explanation:

- A dynamic QR code is generated for each document.
- The QR code links to real-time verification data.
- Users can scan the QR code to verify document authenticity.

7.2.4 AI-Based Validation

```
function analyzeDocument(file) {  
    const text = OCR.extract(file);  
    const analysis = NLP.process(text);  
  
    const score = computeTrustScore(analysis);  
    return score;  
}
```

Explanation:

- OCR extracts text from the document.
- NLP techniques analyze semantic structure.
- A trust score is generated based on detected anomalies.

7.3 Database Implementation

7.3.1 Schema and Structure

The system uses a hybrid storage approach:

- Blockchain stores immutable document hashes and metadata.
- IPFS stores encrypted document files.
- MongoDB stores user data and application metadata.

Table 7.1. Database Tables Overview

Entity	Primary Key	Key Attributes
User	userID	role, wallet address, authentication details
Document	documentID	hash, CID, timestamp, status
Transaction	txID	document hash, timestamp, blockchain record
Verification	verifyID	result, status, timestamp

7.3.2 ER Diagram Reference

The Entity-Relationship diagram for the system is described in the Design chapter. It shows the relationships between User, Document, Blockchain Transaction, and Verification entities.

CHAPTER 8

TESTING

8.1 Functionality Testing

Functionality testing is performed to validate core system operations such as document upload, storage, verification, and QR-based access.

- User authentication and role-based access control.
- Document upload, validation, and encryption process.
- Hash generation using SHA-256.
- Storage of encrypted documents in IPFS.
- Blockchain storage of document hash and metadata.
- Document verification using hash comparison.
- QR code generation and scanning workflow.
- AI-based document analysis and anomaly detection.

8.2 Performance Testing

Performance testing evaluates system responsiveness and efficiency.

- Document upload time and processing latency.
- IPFS storage and retrieval speed.
- Blockchain transaction execution time.
- Verification response time.
- System behavior under multiple concurrent requests.

8.3 Database Integrity Testing

The system ensures data integrity using blockchain and decentralized storage instead of traditional SQL databases.

- Verification of hash consistency between uploaded and stored documents.
- Detection of tampering through hash mismatch.
- Validation of blockchain transaction records.
- Integrity of IPFS-stored encrypted documents.

8.4 Testing Methodologies Used

- Unit Testing
- Integration Testing
- Functional Testing
- Performance Testing
- System Testing
- Security Testing
- Regression Testing
- Manual Testing
- Automation Testing

8.5 Test Environment and Tools

- Operating System: Windows / Linux
- Frontend: React / Next.js
- Backend: Node.js and Express
- Blockchain: Polygon Testnet
- Storage: IPFS

- Database: MongoDB
- Testing Tools: Postman, Jest, Selenium WebDriver
- AI Tools: Python-based OCR and NLP libraries

8.6 Manual Test Cases

Test ID	Test Case	Input	Expected Output	Result
TC-1	Document Upload	Valid document file	Successful upload and storage	Pass
TC-2	Invalid File Upload	Unsupported format	Error message displayed	Pass
TC-3	Document Verification	Original document	Verified as authentic	Pass
TC-4	Tampered Document	Modified document	Detected as tampered	Pass
TC-5	QR Verification	Scan QR code	Redirect to verification page	Pass
TC-6	Blockchain Record	Stored hash	Correct metadata retrieval	Pass

8.7 Automation Testing Approach

Automation testing is used to validate repeated workflows.

- Tool used: Selenium WebDriver
- Scope: UI navigation, document upload, and verification flow
- Execution: Automated scripts with logs and screenshots
- Output: Pass/fail reports for test cases

8.8 Testing Challenges Faced

- Handling asynchronous blockchain transactions.

Table 8.2. Automation Testing Summary

Metric	Count	Remark
Total Test Cases	20	Complete test suite
Passed	19	Stable system
Failed	1	Minor UI delay issue

- Managing IPFS latency during file retrieval.
- Integration complexity between frontend, backend, and blockchain.
- Ensuring consistent AI validation results.

8.9 Performance Testing Results

Table 8.3. Performance Test Results

Metric	Condition	Result
Upload Time	Medium file	Fast response
Verification Time	Normal case	Near real-time
Blockchain Latency	Testnet	Low delay
System Load	Multiple users	Stable performance

8.10 Test Results Summary

- All core functionalities are working correctly.
- Blockchain ensures tamper-proof verification.
- IPFS provides reliable decentralized storage.
- AI module enhances forgery detection capability.
- Minor issues exist in UI synchronization.

8.11 Overall System Readiness

The system is ready for academic demonstration and evaluation. Further improvements can be made for production deployment including scalability optimization and enhanced security measures.

CHAPTER 9

RESULTS AND DISCUSSIONS

9.1 Experimental Setup

The CipherDocs system was evaluated in a controlled development environment to validate its performance, security, and functionality.

The system setup includes:

- Web-based application running on a local development server.
- Frontend developed using React/Next.js.
- Backend services implemented using Node.js and Express.
- Blockchain interaction through Polygon testnet.
- Decentralized storage using IPFS.
- AI validation using OCR and NLP-based analysis tools.

The evaluation focuses on document upload, encryption, blockchain storage, verification accuracy, and AI-assisted validation.

9.2 Parameters Considered

The following parameters were used for evaluation:

- Verification Accuracy
- Processing Time
- Blockchain Transaction Latency
- Storage Efficiency (IPFS CID retrieval)
- AI Validation Effectiveness
- System Reliability under multiple requests

9.3 Observed Results

9.3.1 Functional Results

The system successfully demonstrated all core functionalities:

- Secure document upload with validation.
- Encryption using AES-256 before storage.
- Hash generation using SHA-256.
- Storage of encrypted documents in IPFS with CID generation.
- Blockchain storage of document hash and metadata.
- Real-time document verification using hash comparison.
- Dynamic QR code generation and scanning.
- AI-based anomaly detection in documents.

9.3.2 Performance Results

Table 9.1. Performance Metrics

Metric	Observation	Remark
Upload Time	Fast	Efficient processing
Verification Time	Near real-time	Minimal delay
Blockchain Latency	Low (Testnet)	Acceptable performance
IPFS Retrieval	Reliable	Stable access using CID
AI Processing Time	Moderate	Depends on document size

9.4 Analysis and Interpretation

The results indicate that the system performs efficiently and meets its design objectives:

- Blockchain ensures tamper-proof storage and transparency.
- IPFS provides decentralized and reliable storage.
- Cryptographic hashing successfully detects document tampering.
- Dynamic QR codes enable fast and user-friendly verification.

- AI module enhances detection beyond traditional hash comparison.

The integration of blockchain and AI addresses limitations found in traditional systems, where verification is limited only to hash matching.

9.5 Comparison with Existing Systems

Table 9.2. Comparison with Existing Systems

Feature	Existing Systems	CipherDocs	Improvement
Tamper Detection	Basic	Advanced	Hash + AI validation
Storage	Centralized	Decentralized (IPFS)	Improved security
Verification	Manual / Static	Real-time QR-based	Faster process
Cost	High (Ethereum)	Low (Polygon)	Cost-efficient
Lifecycle Management	Limited	Automated	Smart contracts

9.6 Discussion of Results

The CipherDocs system demonstrates a significant improvement over traditional and existing blockchain-based document verification systems.

- Provides a secure and decentralized approach using blockchain and IPFS.
- Reduces verification time through automated smart contract execution.
- Enhances reliability using cryptographic techniques.
- Improves usability through QR-based real-time verification.
- Introduces intelligent validation using AI-based analysis.

However, certain limitations are observed:

- Dependence on internet connectivity for blockchain interaction.
- AI module performance varies with document complexity.
- Testnet environment may differ from production blockchain performance.

9.7 Final Outcome

The system successfully achieves its objectives of secure document verification, forgery detection, and decentralized storage. It provides a scalable and efficient solution suitable for real-world applications such as academic certificates, legal records, and government documents.

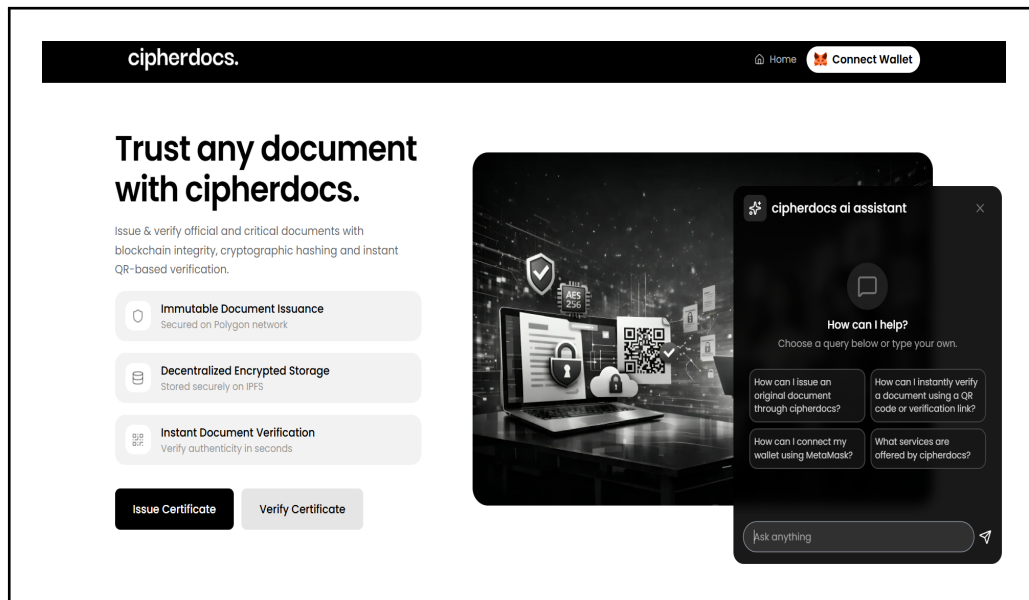


Figure 9.1. Home Page

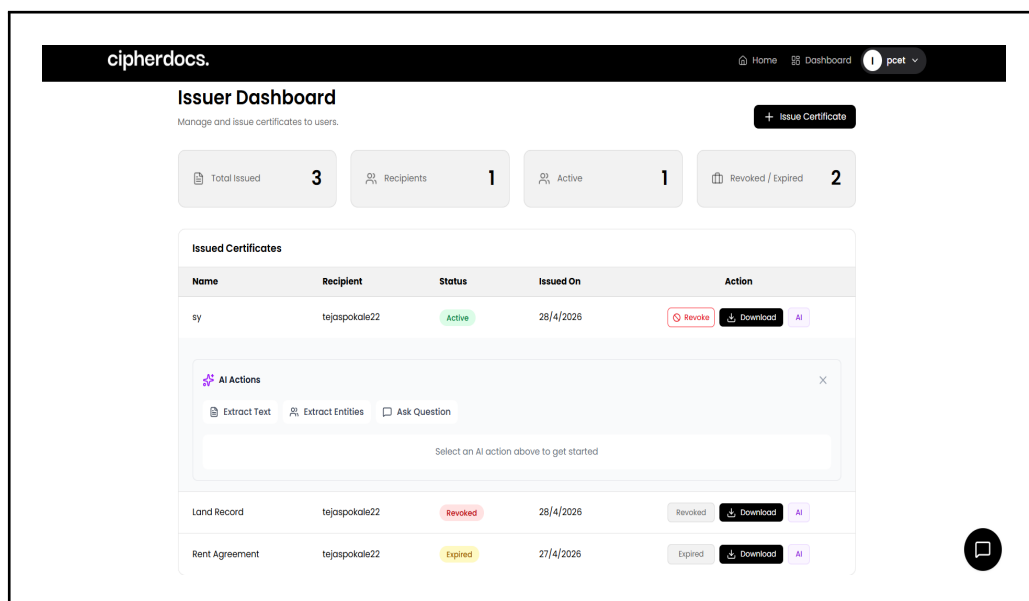


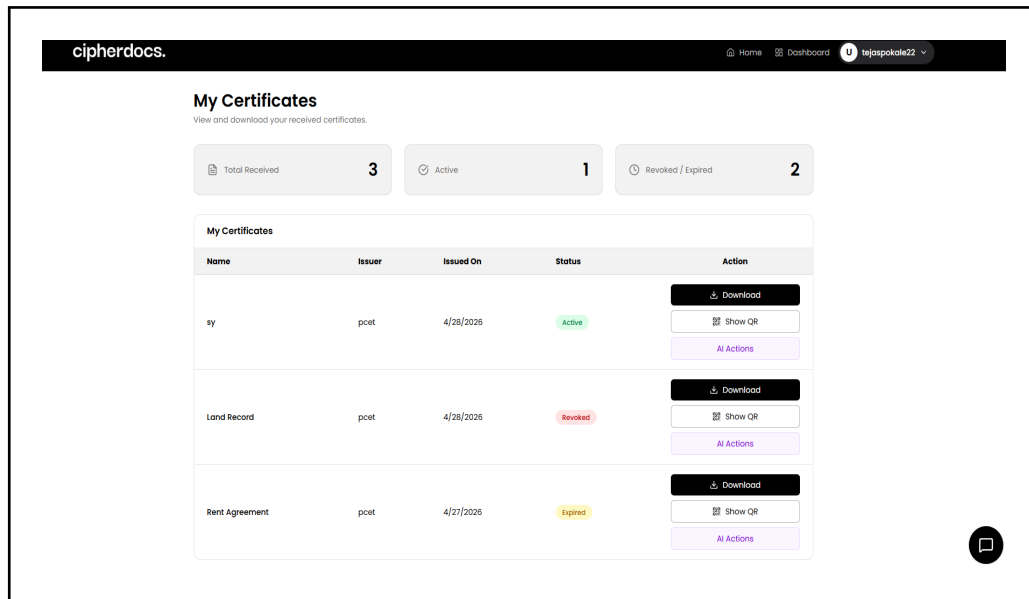
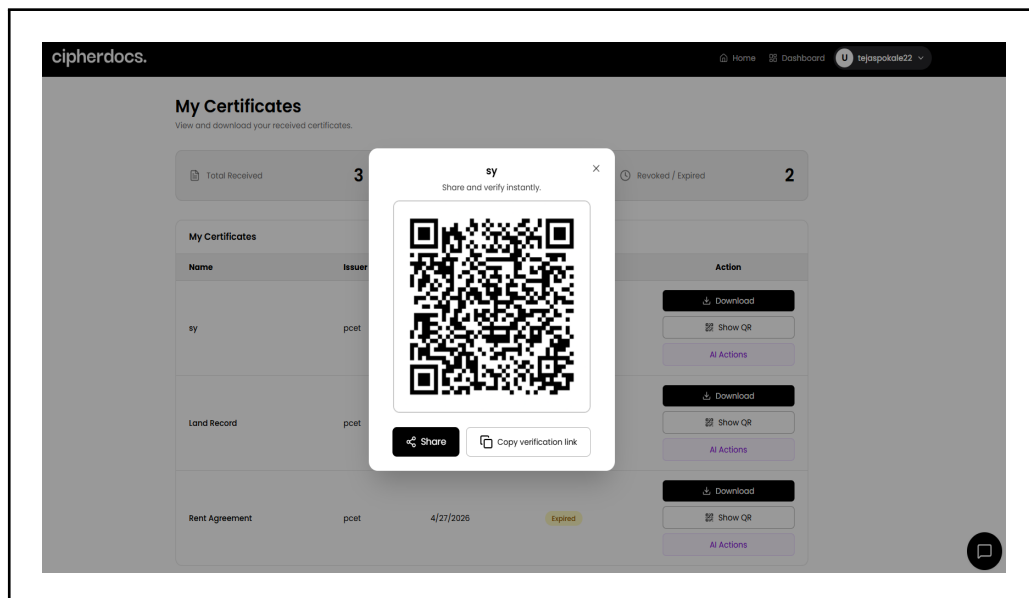
Figure 9.2. Issuer Dashboard

The screenshot shows the 'Issue new Certificate' page on the cipherdocs website. The page has a dark header with the logo 'cipherdocs.' and navigation links for 'Home', 'Dashboard', and a user profile 'pcet'. The main heading is 'Issue new Certificate' with a subtext 'Create and issue a new blockchain-verified certificate.' Below this is a form with four sections: 'Certificate Name*' with a text input field containing 'Enter certificate name'; 'Recipient Wallet Address*' with a text input field containing '0x...'; 'Expiry Date' with a date picker showing 'dd-mm-yyyy' and a note 'Add expiry date only if the certificate has an expiry. Leave blank if it is lifetime.'; and 'Original Certificate Document (PDF)*' with a dashed box for upload and a note 'Click to upload or drag and drop. PDF only (max 10mb)'. At the bottom of the form is a black button labeled 'Issue Certificate'. A small chat icon is visible in the bottom right corner.

Figure 9.3. Certificate Issue Page

The screenshot shows the cipherdocs homepage. The header is dark with the logo 'cipherdocs.' and navigation links for 'Home', 'Dashboard', and a user profile 'pcet'. The main heading is 'Trust any document with cipherdocs.' with a subtext 'Issue & verify official and critical documents with blockchain integrity, cryptographic hashing and instant QR-based verification.' Below this are three feature cards: 'Immutable Document Issuance' (Secured on Polygon network), 'Decentralized Encrypted Storage' (Stored securely on IPFS), and 'Instant Document Verification' (Verify authenticity in seconds). At the bottom of these cards are two buttons: 'Issue Certificate' and 'Verify Certificate'. To the right is a large image of two women working on laptops, with a QR code and a lock icon overlaid. A user profile dropdown menu is open, showing 'PCCOE PUNE Issuer Account', 'Wallet Address' (0xcac8a8144f6e8397795e42278f670fa9c5a721c), a 'Copy' button, and a 'Sign out' button. A small chat icon is visible in the bottom right corner.

Figure 9.4. User Wallet Details

**Figure 9.5.** User Dashboard**Figure 9.6.** Dynamic QR Code

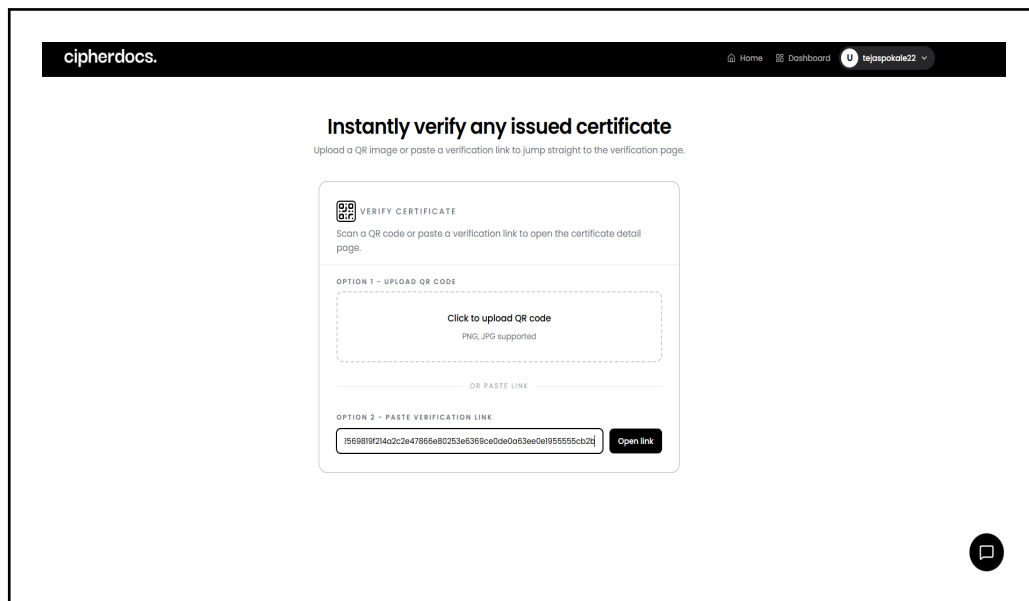


Figure 9.7. Verify Certificate using QR or Link

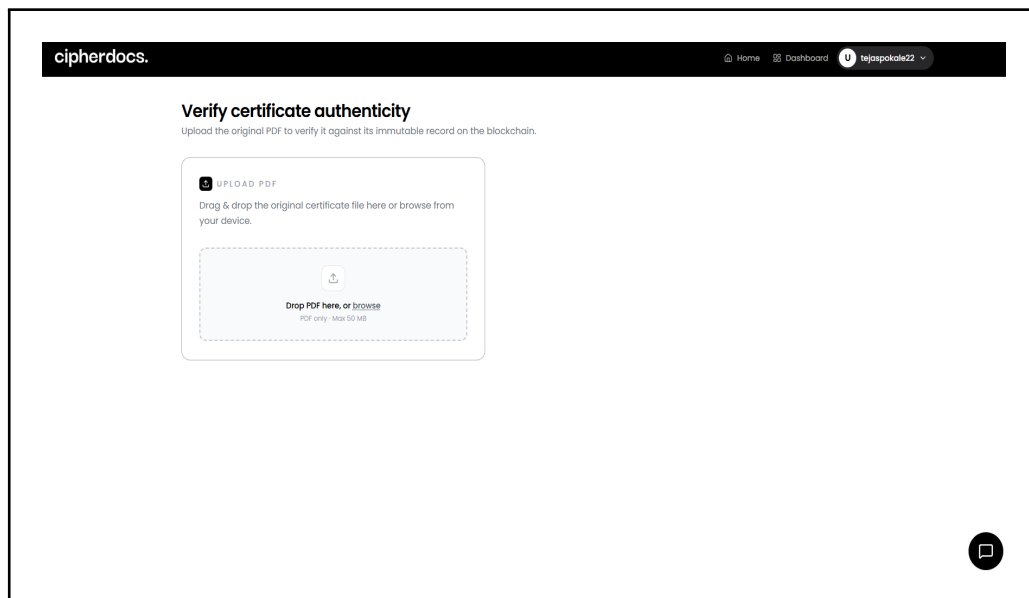


Figure 9.8. Certificate Verification Page

CHAPTER 10

CONCLUSIONS AND FUTURE WORK

10.1 Project Summary

CipherDocs presents a secure and decentralized document verification system that integrates blockchain technology, cryptographic security, and artificial intelligence to prevent document forgery and ensure authenticity.

The system leverages SHA-256 hashing for integrity, AES-256 encryption for confidentiality, IPFS for decentralized storage, and the Polygon blockchain for immutable record keeping. Dynamic QR-based verification enables real-time validation, while smart contracts automate document lifecycle management.

10.2 Key Achievements

The major achievements of the project include:

- Successful implementation of a decentralized document verification system.
- Secure document storage using encryption and IPFS.
- Blockchain-based hash storage ensuring immutability and tamper resistance.
- Real-time verification using dynamic QR codes.
- Integration of AI-based validation for detecting document anomalies.
- Role-based access control for issuers, holders, and verifiers.

10.3 Key Findings

- Blockchain technology effectively ensures data integrity and transparency.
- IPFS provides reliable and decentralized document storage.
- Cryptographic hashing enables accurate detection of document tampering.
- AI-based validation enhances detection beyond traditional hash comparison.
- Polygon blockchain significantly reduces transaction cost and latency compared to Ethereum.

10.4 Challenges Faced and Solutions

- **Blockchain latency:** Managed using Polygon testnet for faster transactions.
- **Integration complexity:** Solved using modular architecture separating frontend, backend, and blockchain layers.
- **Data privacy concerns:** Addressed using encryption and minimal on-chain data storage.
- **AI accuracy limitations:** Improved using OCR and semantic analysis techniques.

10.5 Meeting Project Objectives

Based on the objectives defined in Chapter 1:

- Decentralized document verification system: **Achieved**
- Document integrity using SHA-256 hashing: **Achieved**
- Secure storage using AES-256 encryption: **Achieved**
- Decentralized storage using IPFS: **Achieved**
- Real-time verification using QR codes: **Achieved**
- Smart contract-based lifecycle management: **Achieved**
- AI-assisted document validation: **Partially Achieved**

10.6 Alignment with Sustainable Development Goals (SDGs)

The project contributes to the following SDGs:

- **SDG 9:** Promotes innovation through blockchain-based infrastructure.
- **SDG 12:** Supports paperless and efficient document management.
- **SDG 3:** Enhances reliability of critical documents in healthcare and governance systems.

10.7 Future Work

The system can be further enhanced with the following improvements:

- Implementation of Zero-Knowledge Proof (ZKP) for privacy-preserving verification.
- Deployment on mainnet for real-world scalability.
- Cross-chain interoperability for multi-blockchain support.
- Improved AI models for advanced forgery detection.
- Mobile application integration for wider accessibility.
- Enhanced user interface with multilingual support.

10.8 Applications

The CipherDocs system can be applied in multiple domains:

- Academic certificate verification
- Government document authentication
- Legal and financial record validation
- Healthcare document verification
- Corporate and HR credential management

10.9 Final Remarks

CipherDocs successfully demonstrates a secure, scalable, and cost-effective solution for document verification using modern technologies. By combining blockchain, cryptography, decentralized storage, and artificial intelligence, the system provides a robust framework to prevent forgery and enhance trust in digital ecosystems.

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CHAPTER A

ROUTINE CALCULATIONS AND SUPPORTING MATERIAL

A.1 Standard Calculations

Include all routine calculations performed during project development.

A.2 Standard Data and Tables

Provide reference data and lookup tables used in the project.

A.3 Mathematical Derivations

Document derivations of key formulas or algorithms.

CHAPTER B

CYBER LAWS AND COMPLIANCE

B.1 Applicable Cyber Laws

Document relevant cyber laws and regulations applicable to your project:

- Data Protection Regulations (GDPR, CCPA, etc.)
- Information Technology Act, 2000 (India)
- Intellectual Property Rights
- Software Licensing Laws
- Cybersecurity Standards and Frameworks

B.2 Compliance Measures

Describe how your project complies with identified laws and standards.

CHAPTER C

ADDITIONAL RESOURCES

C.1 Installation and Setup Guide

Provide step-by-step instructions for installing and setting up the project.

C.2 User Manual

Include user documentation and usage instructions.

C.3 Developer Guide

Provide documentation for developers who may maintain or extend the project.

C.4 Configuration Files

Include sample configuration files or environment setup instructions.

CHAPTER D

PLAGIARISM REPORT

Attach plagiarism check report demonstrating originality of the work.

CHAPTER E

PUBLICATIONS AND PATENTS

Attach publication proof and related details. At least one publication is mandatory as per submission requirements.

E.1 Mandatory Publication Evidence

Include the following artifacts:

- Published paper or accepted-paper proof.
- Journal/conference name, paper title, and author list.
- DOI/URL or acceptance letter screenshot.
- Indexing details if available.

CHAPTER F

GUIDE APPROVAL RECORD

Include project guide review and approval proof for the soft copy submitted on or before **31 March 2026**.

F.1 Approval Evidence

- Signed review sheet by project guide.
- Date-stamped approval email/screenshot.
- Revision comments and closure summary.

CHAPTER G

CERTIFICATES AND AWARDS

Include certificates of participation, awards, or recognitions received (if applicable).